Top pre-trained Models for transfer learning

This document lists important pre-trained models across different domains , their use case and where to access them

1. IMAGE CLASSIFICATION

category	Model	Use case	Where to get
CNN	AlexNet	Basic classification	PyTorch Hub
CNN	VGG16 / VGG19	Feature extraction	PyTorch / TensorFlow
CNN	ResNet (18–152)	Deep classification	PyTorch
CNN	Inception v3/v4	Image recognition	TensorFlow Hub
CNN	DenseNet	Efficient feature reuse	PyTorch
CNN	MobileNet v1/v2/v3	Mobile apps	TensorFlow Lite
CNN	EfficientNet (B0–B7)	High accuracy	TensorFlow
CNN	EfficientNetV2	Faster training	google
Transformer	Vision Transformer (ViT)	Transformer-based vision	Hugging Face
Transformer	DeiT	Lightweight ViT	Facebook AI
Transformer	Swin Transformer	Hierarchical vision	Hugging Face
Hybrid	ConvNeXt	Modern CNN alternative	Meta AI
Hybrid	RegNet	Scalable CNN	Facebook AI
Hybrid	NFNet	No normalization	DeepMind

2. OBJECT DETECTION

category	Model	Use case	Where to get
R-CNN	R-CNN	Early detection	GitHub
R-CNN	Fast R-CNN	Faster detection	PyTorch
R-CNN	Fast R-CNN	Region proposals	Detectron2
R-CNN	Mask R-CNN	Instance segmentation	Facebook AI
YOLO	YOLOv1–v8	Real-time detection	Ultralytics
YOLO	YOLOv9 / YOLO-NAS	Latest detection	Deci AI
SSD	SSD	Mobile detection	TensorFlow
Transformer	DETR	End-to-end detection	Facebook AI
Transformer	Deformable DETR	Improved DETR	Hugging Face
Transformer	DINO	Advanced detection	GitHub
Hybrid	EfficientDet	Scalable detection	google

3. FACE RECOGNITION

category	Model	Use case	Where to get
CNN	DeepFace	Face verification	github
CNN	FaceNet	Face embeddings	google
CNN	VGGFace	Face classification	Oxford
CNN	OpenFace	Lightweight recognition	github
Loss-based	ArcFace	High accuracy recognition	InsightFace
Loss-based	CosFace	Face similarity	github
Loss-based	SphereFace	Deep face learning	github
Modern	InsightFace	Industry-level recognition	github
Modern	RetinaFace	Face detection + recognition	github

4. SPEECH RECOGNITION

category	Model	Use case	Where to get
RNN	**DeepSpeech**	**Speech-to-text**	**Mozilla**
Transformer	**Wav2Vec 2.0**	**Speech recognition**	**Hugging Face**
Transformer	**HuBERT**	**Self-supervised speech**	**Facebook**
Transformer	**Whisper**	**Multilingual STT**	**OpenAI**
hybrid	**Conformer**	**Speech + attention**	**Google**

5.COMPUTER VISION

category	Model	Use case	Where to get
Segmentation	**U-Net**	**Medical imaging**	**GitHub**
Segmentation	**SegNet**	**Scene segmentation**	**GitHub**
Segmentation	**DeepLab v3**	**Semantic segmentation**	**TensorFlow**
Segmentation	**SAM**	**Segment anything**	**Meta AI**
Tracking	**SORT**	**Object tracking**	**GitHub**
Tracking	**DeepSORT**	**Tracking with ID**	**GitHub**

6. AUDIO MODELS

category	Model	Use case	Where to get
Audio	**YAMNet**	**Sound classification**	**TensorFlow**
Audio	**VGGish**	**Audio embeddings**	**Google**
Audio	**OpenL3**	**Audio features**	**GitHub**
Audio	**MusicGen**	**Music generation**	**Meta**

7.MULTIMODAL MODELS

category	Model	Use case	Where to get
Vision+Text	CLIP	image-text matching	OpenAI
Vision+Text	BLIP	Captioning	Salesforce
Vision+Text	Flamingo	Few-shot multimodal	DeepMind
Vision+Text	GPT-4V	Vision + reasoning	OpenAI
Audio+Text	Whisper	Speech understanding	OpenAI

8. GENERATIVE MODELS

category	Model	Use case	Where to get
GAN	**DCGAN**	**Image generation**	**PyTorch**
GAN	**StyleGAN**	**Face generation**	**NVIDIA**
GAN	**CycleGAN**	**Image translation**	**github**
Diffusion	**Stable Diffusion**	**Image generation**	**Stability AI**
Diffusion	**DALL·E**	**Text → Image**	**OpenAI**
Diffusion	**Imagen**	**High-quality images**	**google**
Diffusion	**Midjourney**	**Creative images**	**Discord**
Video	**Sora**	**Video generation**	**OpenAI**

9. NLP MODELS

category	Model	Use case	Where to get
LLM	GPT-2	Text generation	openai
LLM	GPT-3	Large generation	openai
LLM	GPT-4	Advanced reasoning	openai
LLM	LLaMA	Open LLM	meta
LLM	Mistral	Efficient LLM	Mistral AI
embedding	Word2Vec	Word vectors	Gensim
embedding	GloVe	NLP basics	Stanford
transformer	BERT	Text classification	Hugging Face
transformer	RoBERTa	Improved BERT	Facebook
transformer	ALBERT	Lightweight BERT	Google
transformer	XLNet	Advanced NLP	Google
transformer	T5	Text-to-text	Google
transformer	BART	Summarization	Facebook
RNN	Seq2Seq	Translation	TensorFlow

10. TABULAR

category	Model	Use case	Where to get
Deep	TabNet	Deep tabular	google
AutoML	AutoGluon	AutomatedML	aws
Tree	catboost	Categorical data	Yandex
Tree	LightGBM	Fastml	Microsoft
Tree	XGBoost	Structured data	XGBoost